Pre-Feasibilty Study

Fruits and Vegetables Grading, Packing Center



2017

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1. Executive summary

The process of grading fruits and vegetables is one of the factors that helps marketing them in the local market and in exportation market and ensure that they reach the consumer as required. The process involves cleaning, sorting and grading according to size, quality and packaging within packages that meet the requirements of the consumer. It also includes the waxing of some of the products in order to save them from damage and store them to be placed in the market at a certain time. The project aims to establish a specialized center to carry out the process of grading vegetables and fruits and the process of waxing vegetables and fruits where it's required targeting the local market and the Arab and European export markets. High quality crops are processed through the provision of post-harvest services including washing, cleaning, sorting, grading, refrigerated warehouses and refrigerated transport. It is possible to start with the available harvest of Jerash and through contracting with farmers or through the purchase from the central markets for trading after providing these services with a capacity of 4,680 tons of different types of vegetables and fruits that need to be processed and packaged.

The following tables shows the main results of the study:

Table 1 project main results

Project name	Fruits and vegetables Grading and Packing Center
Proposed location	Jerash
Proposed service	Washing, sorting, grading, packing
Total labor (person)	17
Total investment (thousand JD)	504
Net present value of cash flows (thousand JD)	3,914
Pay back period (years)	2
IRR	%55.26

The estimated capital cost of the project is JD 504,309, mostly in construction (89,375), land (110,000), machinery and equipment (140,305), vehicles and machinery (44,000) and working capital which represents the largest share due to the hypothesis of trade in vegetables and fruits after processing (133,279), Where working capital is required for production inputs and workers' salaries. The project will be 50% financed through loans and the rest through self financing. Despite the high investment cost of the project, it will provides suitable job opportunities commensurate with the size of the investment.

The project will rely on exploitation of untapped raw materials and relying on labor in manufacturing and marketing Processing and filling different kinds of fruits and vegetables.

The results of the above financial analysis indicates that the net real value of the project's cash flows is positive, which is an indication of the financial feasibility of the project. The internal rate of revenue is much higher than the weighted average cost of capital which is an indication of the feasibility of the project. Other financial indicators indicate that the project is profitable for the investor.

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

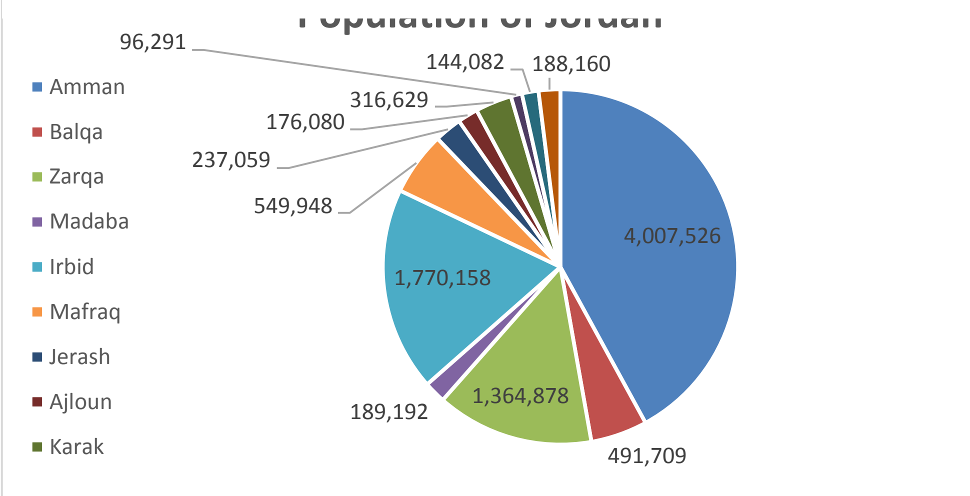
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

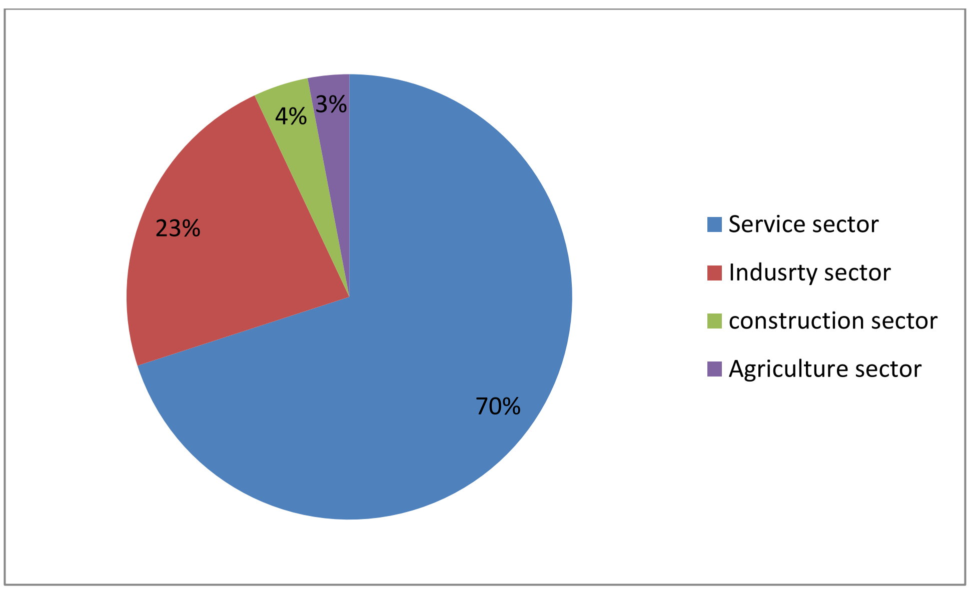
Department of Statistics (DOS), 2015.

Overiew of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the

Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market analysis

Project description

Factors affecting the purchase of horticultural products vary in different markets depending on the nature of the buyer (consumer) in that market. However, buying motivation remains very similar to the different markets. Socio-economic factors also plays an important role in determining the required type of goods. Whatever the factor, the main factor is quality in the first class and price in second class. The main factor in this field is the quality of the product in terms of being fresh and has excellent qualities and good packaging. The quality factor is followed by price especially for high income categories.

The data indicates that the demand for fresh horticultural products is increasing continuously, especially for some varieties of fresh vegetables, fruist, organic products and re-packaging products, especially salads and green paper products, which constitute one third of the expenditure on vegetables. One of the most important weaknesses of the fruit and vegetable sector is the lack of applied experience in the field of appropriate practices before and after the harvest in addition to the correct marketing practices, including the weak infrastructure of marketing for horticultural products, especially in regard to sorting and grading and initial cooling, packaging, refrigerated stores, Transportation, etc.

Freshly harvested crops are prepared by multiple processes to be ready for marketing, shipping or caching, the main processes are preparation, sorting, cleaning, disease control, drying, industrial ripening, reveal flower buds, prolonging flower life, polishing, grading, packing, But it's not necessarily that all crops go through all these processes, there are crops that do not need to be prepared and others do not need to be ripen and so on. These operations are carried out in the house of the plastic houses or in special houses, so that the freshly consumed crops have several processesin preparing them for marketing and shipping or temporary storage. The most important of these operations are: preparation, sorting, cleaning, disease control, drying, industrial maturation, polishing, Packing, initial cooling. But it is not necessarily that all crops go through all these processes, there are crops that do not need to be prepared and others do not need to ripen and so on. These operations are carried out in a specialized center

Project objectives

There is an increasing demand on fresh fruits and vegetables packaged in small and homogeneous weights, as a result of the change in the consumption pattern of the community and the increasing rates of working womens outside the house. The fruits and vegetables markets suffer from the heterogeneity of the products, and in their gradualization and packaging. The project aims to produce a variety of vegetables and fruits, and provide the service of cleaning, packaging, transporting and marketing in small bags or packages suitable for the consumer, especially for segments of high income who wish to pay for a clean and fresh products. The project aims to provide the preparation of the products for marketing in the local market and for export ., ensuring the use of technologys to achieve high quality and reduce costs. These services include: initial cooling, cleaning and peeling, grading and packaging, refrigerated transport, refrigerated storage, in addition to using the best packaging quality, that helps in promotion purposes, and add the appropriate statement card to different markets and hypermarkets.

The projects aims:

1. Achieving high profit for the project owner.
2. Providing product development services for marketing in the local market and for export, ensuring the use of technology to achieve high quality and reduce costs.
3. Supply vegetables and fruits after adding the functions of the value chain to them for different the types and supply them to the local market with distinctive good quality.
4. Prepare products with better specifications and packaging suitable for high income segment.
5. Meet part of the local demand for vegetables and fruits that are graded, homogeneous, chilled and ready for cooking
6. Introducing new types and varieties required by the markets, especially those that achieve a high return, or what Jordan produces with a comparative advantage
7. creating job opportunities In the less developed regions.

Such projects are considered important projects and can be considered pilot projects that require traning the staff and labors before starting the project. This project can open up new the employment prospects. This project does not require agricultural land or large areas. The surplus of vegetables and fruits can be exploited them in order to export them to the foreign markets.

Proposed product

In most agricultural seasons and at the peak of production, the production of fruit flows into the market, causing a greater supply for the material in the markets, causing a sharp drop in prices. This is often happened repeated ly in most seasons; this phenomenon will be exacerbated in the coming years as a result of the expansion of irrigated agriculture in the highlands, especially in Mafraq and Arzaq .

There are many constraints and artistic parameters that faces horticultural sector in Jordan including the lack of commitment of the suppliers to the minimum quality of the products, where the technical reports indicated a significant variation inthe quality of the products as goods of different sizes and quality reach themarkets in the same package, which requires a greater effort from the retailer to resort the goods before offering them to the customers , which raises the cost due to the high rate of loss and damage and consequent increase in prices of the reason of the variation in volume products is due to the limited efforts of the producers and the guiding method (by placing good and excellent fruits on the top of the package, and the lower quality products or even corrupted at the bottom of the package), and because there are a large number of small producers and commissioners, which make it difficult to comply to technical standards and other technical difficulties are also that the incoming shipments to the market contain packages of different sizes and with more than one quality and from different geographic area or source with different specifications in terms of quality, shape, maturity and packing in the same shipment causing difficulties for customers to distinguish the product in the market .

A study that was prepared at the University of Jordan for the Ministry of Agriculture, it included a study of the behavior of consumers about the preferred place of purchase and the appropriate quality and the current packing of vegetables and fruits and the impact of income level and the number of family members on the amount of purchase and the desire to buy ready packed products packed and what weights preferred and how much desire to buy graded products, and how much do you want to pay for graded product? What is the size of the packagings that the consumer wants to buy, What is the diffrence in price that the consumer can pay in the case of a homogeneous product.

The results showed that more than half (54%) of consumers said that the marketing of vegetables and fruits in their current quality and packaging is not appropriate, and that the percentage of consumer purchases from potential sources of purchasing vegetables and fruits is about 59% of total consumer purchases from retailers (vegetable shops), 10% of supermarkets, and 11% of street vendors, 43% of consumers expressed a desire to buy ready-made packaged products with specific weights. There was a disagreed and varied opinions on the size that consumers prefer to buy. The results showed that 58.2% of consumers wants to purchase packed product with specific weights and preferred to packs with a weight less than 2 kg, it was found in the sample of the study about their preference for the method of choosing the fruits in their hands when buying vegetables and fruits, 94% of them preferred the method of choosing the fruits and vegetables by their hands when purchasing them to ensure their validity and the absence of defects, 6% of consumers don't prefer this way of buying. The average percentage of consumers wishing to buy the first class high quality, graded, packed fruits and vegetables were 53% of the total consumer desires, while the percentage of the second class was 38% and 73% of the respondents Intended to pay a higher price for graded homogeneous, non-counterfeit products,

The proposed products for the project includes: Graduation and sorting of vegetables and fruits in small, homogeneous containers, such as tomatoes, cucumbers, green beans, carrots, green peas, carrots, okra, spinach, grape leaves, green beans,cabbage, oranges and lemons,and distinctive products of low demand suc as broccoli and artichocke. Other items that are suitable for graduation.

In addition to the vegetable and fruit grading, the project will prepare fresh vegetables that are ready for cooking and packaged in plastic bags (Polyethylene, trays or cardboard dishes) sealed with uniform weights for each type. And the following are an examples for the provided services:

1. Preparation of the molokhia and equip them in 500-1000 gm packages.
2. Preparation of zucchini and aubergine.
3. Preparation of ingredients of vegetable soup in 500-1000 g plate
4. Preparation of garlic in plastic bags with capacity of -250 g
5. Preparation of onions and graduation it in a plastic bags 1-2 kg
6 - Processing spinach and leaves of vegetables and placing in polisterin cartons 0.5/1 kg
7. Processing of all kinds of green leaves of vegetables (mint, parsley, coriander), processed, cleaned and packaged in plastic trays covered with plastic rap, capacity 1 / 0.5 kg
8. Processing of zucchini and aubergine in homogeneous sizes and packaging in plastic trays covered with plastic rap capacity 1 / 0.5 kg
9. Graduation and filling of various types of vegetables in homogeneous sizes and homogeneous maturity and packaging in plastic trays coated with plastic capacity 1 / 0.5 kg

The packing will be carried out by a trained and expert workers without any contact between the worker's hand and the fruits. In order to comply with the health conditions, the hygiene is strictly observed in the production department so that all the workers are following strict hygiene procedures, to ensure the quality of the products and to keep them fresh, there are high efficiency refrigerators to protect the products before and after processing.

Market size analysis

Jordan has made significant progress in increasing agricultural production during the past decade. Plant production has doubled since 2000, with agricultural output increasing from 1.5 million tons in 2000 to 3 million tonnes in 2015. (Table 4): where the production of vegetables increased from 966 tons in 2000 to about 2 million tons in 2015. This achievement was achieved mainly by the expansion of irrigation networks, the expansion of the use of green houses, and using high productivity hybrid seeds. One of the reasons for the increase in production was also the response to the expansion of local and international demand for fresh products in foreign markets. However, this expansion in production has not been accompanied by major development in crop handling techniques and post-harvest treatments.

Table 3 Development of agricultural production during (2000-2015) per ton

The crop	2000	2005	2010	2015
Grains	60,421	108,632	92,001	98,217
Field Crops	65,428	266,056	225,527	250,795
Winter vegetables	462,896	837,848	931,884	1,166,268
summer vegetables	503,112	722,917	858,253	881,167
Total vegetables	966,007	1,560,764	1,790,137	2,047,435
Tomato	354,292	598,933	737,261	870,017
citrus fruits	124,595	136,282	118,632	125,841
Olive	134,285	113,070	171,672	200,896
fruits	112,406	159,372	169,938	294,413
Fruits and vegetables except olive	1,203,009	1,856,418	2,078,707	2,467,689
Total	1,463,143	2,344,175	2,567,906	3,017,596

The previous table shows that the tomato crop is the dominant vegetable crop during the previous period followed by potatoes and watermelons, which are produced in both the highlands and the Jordan Valley. The following table shows the quantities of fruits and vegetables produced during the period (2011-2015), showing clearly the increase in annual quantities of vegetables and fruits (Table 4).

Table 4 Evolution of production quantities of vegetables and fruits during (2011-2015) ton

The crop	2011	2012	2013	2014	2015
Winter vegetables	1,132,144	899,687	924,867	990,661	1,166,268
summer vegetables	796,159	808,047	920,359	939,415	881,167
Total vegetables	1,928,303	1,707,734	1,845,225	1,930,076	2,047,435
citrus fruits	106,848	111,080	96,261	94,204	125,841
Fruits(except olive)	187,832	188,410	211,822	204,257	294,413
Total	2,222,982	2,007,224	2,153,308	2,228,536	2,467,689

These quantities need infrastructure, marketing, advanced grading and packaging services, and an increase in agricultural production. The main justification for the need for infrastructure stems from the fact that Jordanian producers and exporters and all stakeholders must comply with international regulations and standards if they wish to compete in foreign markets. Therefore, maintaining an advanced position in this competition requires greater consideration of quality assurance in all post-harvest steps. The low cost of production in addition to trained workforce will also increase competition in the sector.

The data of foreign trade in the Department of Statistics indicate that there is relative stability in the quantities of vegetables that are imported and exported during the period (2011-2015), which are classified under the seventh chapter (07) and come under a number of tariff clauses. The imported quantity from vegetables is about 185 thousand tons while the amount of exports about 750 thousand tons that worth about 360 million JD, and if the post-harvest processes for these exported goods it would have doubled the amount of exports, as shown in the following table (Table 5).

Table 5 Imports and Exports of Fresh Vegetables during the Period (2011-2015)

year	Imports (thousand JD)	Imports (tons)	Exports (thousand JD)	Exports (tons)	Re-exported (thousand JD)	Re-exported (tons)	total exports (thousand JD)	Trade balance (thousand JD)
2011	84,205	178,615	336,780	755,489	3,484	9,276	340,264	256,059
2012	81,028	179,042	330,907	688,681	3,570	9,987	334,477	253,449
2013	90,422	201,522	335,121	813,713	2,881	7,311	338,002	247,580
2014	84,996	212,521	427,558	766,749	3,822	11,820	431,379	346,383
2015	76,073	154,408	370,896	650,130	2,802	6,806	373,698	297,625
Average	83,345	185,222	360,253	734,952	3,312	9,040	363,564	280,219

The average of exports of fruits (HS 08) was around 115 thousand tons, the rest was about 117 million JD (Table 6). The total rate of Jordan's export of vegetables and fruits was about 850 thousand tons valued at 480 million JD for the period (2011-2015) as shown in Table (7). The Arab Gulf markets are considered a traditional markets for Jordanian products. These markets accommodates about 97% of the total exports of vegetables and fruits and the remaining 3% are to non-Arab markets, 2% of which are to Western and Eastern Europe and 1% for other countries.

In terms of the quality of exports to the Gulf markets and neighboring countries, most of them are similar to the quality consumed in the local markets, where large quantities of these exported products are imported to the three main wholesale markets in Jordan and are exported by refrigerated vehicles to the Gulf markets, and some products are exported in ordinary to neighboring countries(Iraq, Syria and Lebanon.) The main competitors in the GCC markets are Lebanon, Turkey, Egypt,

Table 6 Imports and Exports of Fresh Fruits during the Period (2011-2015)

year	Imports (thousand JD)	Imports (thousand tons)	Exports (thousand JD)	Exports (thousand tons)	Re-exported (thousand JD)	Re-exported (thousand tons)	total exports (thousand JD)	Trade balance (thousand JD)
2011	136,693	177,033	85,392	82,504	4,504	5,631	89,895	-46,798
2012	151,567	179,940	126,821	109,363	4,727	5,262	131,548	-20,019
2013	168,956	193,457	135,914	132,366	5,076	4,745	140,990	-27,966
2014	181,489	211,786	103,851	121,588	4,014	5,954	107,865	-73,624
2015	226,149	204,955	135,752	133,812	3,036	3,725	138,788	-87,360
Average	172,971	193,434	117,546	115,927	4,271	5,063	121,817	-51,153

Source: department of statistics

Table 7 Imports and Exports of Vegetables and Fresh Fruits for the Period (2011-2015)

year	Imports (thousand JD)	Imports (thousand tons)	Exports (thousand JD)	Exports (thousand tons)	Re-exported (thousand JD)	Re-exported (thousand tons)	total exports (thousand JD)	Trade balance (thousand JD)
2011	220,898	355,648	422,172	837,993	7,988	14,907	430,159	209,261
2012	232,595	358,981	457,729	798,044	8,297	15,249	466,026	233,431
2013	259,378	394,979	471,035	946,079	7,957	12,056	478,992	219,614
2014	266,485	424,307	531,409	888,337	7,835	17,774	539,244	272,759
2015	302,222	359,363	506,649	783,942	5,838	10,531	512,487	210,265
Average	256,316	378,656	477,799	850,879	7,583	14,103	485,382	229,066

Source: department of statistics

The net domestic demand for vegetables and fruits (excluding olives) can be calculated by knowing the quantities of domestic production at the expense of imported, exported or re-exported quantities for the last four years (2011-2015). Table (8) shows the amount of domestic demand for vegetables and fruits.

Local market production and consumption of fruits and vegetables was estimated as shown in (Table 9) based on the net amount of imports rate minus exports and adding local production that is estimated about 2.4 million tons, and that per capita consumption of vegetables and fruit rate amounted to about (212) kg per year, assuming that the consumption rate per capita of vegetables and fruits will remain in the same limits of population growth and given that the rate of consumption is linked to the number of population as a result of increased demand on food, and where the rate of population growth in the Kingdom, according to the Department of Statistics is 2.2% per year, and assuming that Assuming that the demand growth rate is equal to the population growth rate. The predicted local demand on vegetables and fruits of all kinds during the period (2018-2027) is shown in table(7)

Table 8 Evolution of Imports and local Production of Vegetables and Fruits during the Period (2011-2017)

Jordan market size from fruits and vegetables	2011	2012	2013	2014	2015	2016	2017*
Imports (tons)	355,648	358,981	394,979	424,307	359,363	1,167	1,190
Exports (tons)	837,993	798,044	946,079	888,337	783,942	807,460	831,684
Re-exported (tons)	14,907	15,249	12,056	17,774	10,531	10,847	11,173
Local production (tons)	2,222,982	2,007,224	2,153,308	2,228,536	2,467,689	2,517,043	2,567,384
Total demand on the product (tons)	**1,725,730**	**1,552,912**	**1,590,152**	**1,746,733**	**2,032,579**	**2,073,231**	**2,114,695**

Source: Department of Statistics (2015), (2) Central Bank (2015)

Table 9 Expected demand for vegetables and fruits during the period (2018-2027)

expected demand on the product	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Expected population (thousand person)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Annual per capita consumption (kg / person)	212.6	212.6	212.6	212.6	212.6	212.6	212.6	212.6	212.6	212.6
Expected demand on vegetables and fruits (thousand tons)	2,122	2,169	2,217	2,266	2,315	2,366	2,418	2,472	2,526	2,582
Expected Project Production (Ton)	2,340	4,680	4,680	4,680	4,680	4,680	4,680	4,680	4,680	4,680
Project market share of the local demand (%)	0.11%	0.22%	0.21%	0.21%	0.20%	0.20%	0.19%	0.19%	0.19%	0.18%

Source: Study Group Preparation

Competitors analysis

Horticultural products are marketed in the local market through traditional marketing methods with relatively high costs and margins compared to their added economic benefits through these channels. As it does not add any economic benefits due to the lack of any marketing, such as the initial cooling and gradualization and packaging and appropriate packaging and cold storage and processing, which reduces the added value and job opportunities that can be created by these processes and reduces the marketing life of products and quality and increase the rate of loss and damage during trading. All specialized studies indicate that the current situation of marketing of horticultural crops weakens their competitiveness in the local and external markets and missing many available marketing opportunities.

There are eight wholesale markets in Jordan, one of which is in Amman and the others are in Irbid, Zarqa, Karak, Mafraq, Tafila, Sult, Aqaba and Jerash. Most of the vegetables and fruit produced in Jordan are imported through the Central market of Amman. Most of the exports are also purchased from the same market. The volume of vegetables and fruits received in the three main central markets reached 913 thousand tons in 2016. The Central market of Amman deals with 626 thousand tons annualy, Irbid central market deals with 215 thousand tons and the market of Zarqa about 36 thousand tons. These quantities constitute the majority of marketed vegetables and fruits in the Eight Wholesale markets.

There are fifteen grading and packaging stations owned by private and public sectors that have been established in Jordan since the 1970s. Some of these stations are equipped with the advanced technology of grading, sorting, waxing, drying and packing. The movable units are used for cranes, loading and unloading. These filling stations have refrigerated warehouses, primary cooling units and ripening rooms. While some other stations are semi-automatic or manual. There are two main centers in Jordan Valley owned by Sunfruit International company , purchased from the Jordanian Company for Marketing and Manufacturing Agricultural Products, one of which is in the South Shouneh and the other in the area of Wadi Al-Yabes .

Most of the current grading and packaging stations do not perform the expected functions efficiently in promoting high-quality products for exportation. The main constraints in the system of grading and packaging stations in Jordan includes: 1) mostly owned by the private sector and serving their interests only 2) mostly are dedicated for specific crops in large farms; 3) insufficient or nonexistent in some production areas 4) Machines doesn’t exist, so their production capacity is low 5) some grading and filling stations do not work.

Table10 list of grading and packing stations in Jordan

Grading and packing center name	location	Production rate(ton/hour)
Sun Fruit International Station	South shuna	30
Sun Fruit International Station	Wadi Al yabes	30
Jordan Valley Foundation Station (JOVAC)	Middle ghor	5
Shrap station	Al-jweideh	15
the south fridge grading station for the apple	Al-qastal	5
Mashreq plant for vegetable processing	Dir- alla	5
AFCO company for the gradualization of vegetables	Al- yadudah	5
Abu Al-Hajj Plant for apple processing	Shoubak	5
Alayyan and Hashlamoun station for apple processing	Shoubak	5
Znouna plant for apple processing	Shoubak	5
Shaheen Sons Station in (not working)	Middle ghor	-
Ibrahim Sheikh packing station (not working)	-	-
Agrico Filling Station (not working)	-	-
Fresh Fruits Company	Amman	22
Abdul Hamid Al Hashlamoun Station for apple grading	Shoubak	5

Various types of containers are used in Jordan for the local market and exportation markets. Polystyrene containers dominate local and Gulf markets, this type of boxes is weak, and does not provideright vertical support. Polystyrene boxes are heat-insulating, environmentally harmful, and aren't recyclable. Various types of packaging are used in Jordan for the local market and exportation markets. The types that are currently used are made of polystyrene, carton, plastic and wood. There are eight factories for the production of polystyrene containers in different locations

Data from the Association of Vegetable and Fruit Exporters indicates that there are 148 exporters with an annual export rate of 3,429 tons per tonne per month. These data show that 30 exporters export 70% of the total Jordanian exports. Their main locations are located in Abu Alanda, Al Rajeeb, Sahab, Juwaida, Wihdat / Amman. These exporters work in various other commercial functions in addition to export as producers, intermediaries, wholesalers, importers, etc. Their workshops lack any kind of machines and equipments needed to prepare crops for markets and are limited to simple manual operations. Sometimes adhesive tapes are added to packaging or packed and loaded in refrigerators. But the predominant nature of the work of these workshops is to load the refrigerators directly from the small vehicles coming from the farm without any processes at all. All kinds of packagings are used (wooden, polystyrene, plastic, carton).

There are 18 exporters to European markets who prepare crops for export in grading and packaging stations, such as sorting, grading and packing using modern equipment. Quality control on the product is necessary in these plants to comply with international requirements.

Prices analysis and pricing strategies

Postharvest transactions are important nodes in the Value Chains of agricultural products of all types, and are more important in horticultural crops of vegetables and fruits, due to the positive or negative effects of good or poor postharvest practices on those Products. It is defined as the processes applied to the agricultural product from the start of the harvest or harvest date to end up for sale and consumption. This includes the group of post-harvest agricultural practices in the field and the range of operations that fall within the framework of marketing practices such as refrigeration, storage, packaging and transport. Thus, economic and commercial characterization has dominated the processes involved in the post-harvest series, both in terms of persons types and institutions that carry out the work, or in terms of the results and objectives to be achieved by good application, or by the results that can be achieved for farmers, sorting and grading centers, trasporters, wholesalers, retail and consumer.

The most important returns that the project earns at all stages of production and trading through the good application of post-harvest transactions can be summarized as follows:

- Reduce the losses of between 20 - 40%.
- Increase the competitiveness of products in front of their counterparts in the markets and opportunities to sell and thus increase income.

Despite the weakness of marketing and the limited provided services by various intermediariesby means of these channels,themarketing margins are unjustifiably high. The difference between the farm price and the consumer price in most of products is high, and that indicates the high profits for intermediaries of the marketing channels at the expense of the product and the consumer, and the high rate of loss and damage, which represents a clear lack of efficiency of the marketing .

The system of sale declared in the markets is the auction system according to the law of municipalities in 1955, with the aim of achieving the greatest justice possible to the parties of the process by facing the supply demand and create the atmosphere of competition
. However, due to the weakness of regulation system of sales in the market, the dispersion of the auction weakens this goal and turns many sub-auctions into direct salesabout the price between the seller and the buyer. This is due to the fact that there are hundreds of sub-auctions that take place at the same moment in the same market, thus dispersing buyers among the stacks of vegetables and fruits. However, the bulk of the purchases are made by exporters who handle large quantities of hundreds of tons per day. Table (11) shows the price of FOB for Jordanian exports from Amman Customs Center. Noticing the reduction in the average price of exports compared with the retail prices for the Jordanian consumer.

Table 11 Exports average prices of vegetables and fruits (JD/ton)

Year	vegetables	Fruits	Proposed price
2011	446	1,035	504
2012	480	1,160	574
2013	412	1,027	498
2014	558	854	598
2015	570	1,015	646
Average	490	1,014	562

Expected market share

The project targets exporters and residents with high incomes and rich people of the regions who buy to satisfy their physical and psychological needs together so that the price does not intervene in their purchase decision, but quality occupies the first place among the elements of the purchase decision. The quality of agricultural products is much more important than the price. Thisidea is in line with the markets in developed Western countries where consumerslook for quality first in comparative tothe price that is in the fourth or fifth degree when taking decisions, after quality come other properties such as appearance, packaging, nutritional value, Safe on health, and fresh.

In general, farmers and consumers are not accustomed to the quality requirements of used materials and the controls of packaging, numbering, as wellas,producers and exporters are not sufficiently aware of the significant gains that can be obtained from the high quality products, as in this case the product marketed itself rather than offering it for sale. There is a big chance for Jordanian exporters to enter strongly in the Gulf markets, which require high-quality products with a distinctive packagingof fruits and vegetables (such as sherry tomato and clusterd tomatoes, sweet peppers of different colors, small Asian chili peppers, bean , cauliflower, Cowbeans,local lettuce , Iceberg, strawberries, seedless grapes, peaches and nectarines)

The proposed market share of the project is a modest quota of 4,680 tons per year, i.e a rate of grading and sorting about 15 tons per day. As the production quantities in Jerash is not large compared to other areas,they do not exceed 0.15% of the total demand for vegetables and fruits.

It is not expected that the project will face difficulties in achieving expected revenues due to the absence of a projects offering similar services in the region. Such a project would help fruits and vegetables farmers to market their products in suitable quantities and prices in the local and foreign markets in the event of providing services instead of trading.

The average price for grading vegetables and fruits, or sorting and polishing of citrus fruits, varies between 120-150 dinars per ton. For the purposes of this study, the project is assumed to trade vegetables and fruits by buying them from farmers or wholesale markets based on the local prices and adding post-harvest services. Prices should include the required packages

Expected revenues analysis

The project is expected to produce an annual average of 4,680 tons per year of vegetables and fruits that are classified, graded and packaged in packages suitable for each category ,at a rate of 15 tons per day, knowing that there are seasons of production and may have to stop in some periods of the year especially in winter days.

For the purposes of this study, it was considered that the weighted average price for the selling price is (650) dinars /ton , after the operations of processing and packing various vegetables, note that the weighted average purchase of raw materials from vegetables before sorting and gradualization amounted to about (319) JD per tonne, where for fruits (547) JD per tonne and for all fruit and vegetable products amounted to about (387.4) dinars per tonne as a weighted price based on the quantities received for the three central markets and most of the prevailing price for 2016. The selling prices of vegetables and fruits of all kinds is expected to increase by 2% per year. Table (Table 12) shows the total expected revenues.

Table 12 Expected selling prices and total sales during the period (2018-2027)

Project revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2018
Total sales (kg)	2,340	4,680	4,680	4,680	4,680	4,680	4,680	4,680	4,680	4,680
Product price(JD/kg)	650.0	663.0	676.3	689.8	703.6	717.7	732.0	746.6	761.6	776.8
Total sales (thousand JD)	1,521.0	3,102.8	3,164.9	3,228.2	3,292.8	3,358.6	3,425.8	3,494.3	3,564.2	3,635.5
Total revenues (thousand JD)	1,521	3,103	3,165	3,228	3,293	3,359	3,426	3,494	3,564	3,635

4. Technical analysis

The fruits and vegetables processing industry is characterized by the diversity of its products and thus the multiplicity of production lines, while some facilities are limited to one or two lines of production, and other facilities includes all production lines. The service and assistance units provide water and energy supplies to the facility and requirements of maintenance, storage, packaging, testing and quality analysis. Regarding the exposure of the raw materials of fruits and vegetables to microbial contamination, the design of the required machinery and equipment should allow easy cleaning and sterilization and achieves sanitary conditions during operation.

Required raw materials

In order to produce the required quantity which is 4,680 tons, it is necessary to purchase raw materials of vegetables and fruits in a quantity estimated about (5382) tons after taking into account that the average of the waste and the damage is 15%, noticing that some of the items have a much higher percentage and some have less. Assuming they were purchased from the central markets at the market prices as shown in Table (13) and Table (14), where the estimated price rate of vegetables (319) dinars per tonne, and the fruit is (547) dinars per tonne, and for all fruits and vegetables products average prices are about 387.4 dinars per tonne based on the quantities received form the three central markets. The figures in table (13) and table (14) were used in estimating the cost of required raw materials. The estimated cost of raw materials of vegetables and fruits is (2,084,774) JD, and the project needs packaging materials and packaging cardboard and detergents and disinfectants and chemicals with an estimated amount of (250,000) JD annually.

30

Table 13 Quantities and wholesale prices of vegetables received to the central markets

Type	Zarqa market		Irbid market		Amman market		Central markets	
	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price
Aubergine	1,226	199			24,826	221	26,052	210
Thin eggplant	511	273	534	306	10,155		11,200	289
Classic Eggplant			12,519	219	7,588	356	20,107	288
Pea	67	726	81	560	755		903	643
Green Okra	130	2,081	993	1,899	1,887	595	3,010	1,525
Green onion	242	624	453	499	2,813		3,508	561
Dried onion	3,138	345	13,738	348	59,196	314	76,072	336
Potato	3,731	370	36,966	359	99,514	381	140,211	370
Parsley	780	81	6,169	115	6,580		13,529	98
Tomato	7,594	193	63,733	198	110,462	306	181,789	232
Green garlic			287	547	1,145		1,432	547
Dried garlic	521	1,635	1,419	1,822	3,161	1,933	5,101	1,797
Carrot	546	368	4,872	335	16,571	374	21,989	359
lettuce	434	185	1,421	179	6,483		8,338	182
Cucumber	2,616	315	26,183	316	72,674	331	101,473	321
yellow corn	2,761	380	4,808	309	11,198		18,767	344

Type	Zarqa market		Irbid market		Amman market		Central markets	
	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price
Thyme	94	768	35	751	965		1,094	760
cauliflower	2,478	267	12,129	269	33,763	314	48,370	284
Yellow cauliflower	20	400			50		70	400
spinach	895	222	3,467	200	10,240	177	14,602	200
beet	28	410	56	521	892		976	465
Schumer			198	311	569		767	311
bean			243	895	5,240	911	5,483	903
Green bean	424	966	1,108	796	43		1,575	881
radish	471	228	2,764	230	5,638		8,873	229
Wild cucumber	65	385	166	629	1,896		2,127	507
Hot pepper	790	436	3,538	349	11,225	424	15,553	403
sweet pepper	886	401	6,808	327	19,350	392	27,044	374
Green Bean	246	639	1,502	664	4,292	425	6,040	576
pumpkin	23	350			549		572	350
Zucchini	1,300	342	9,347	371	26,968	428	37,615	380
kale	166	315	484	267	2,122		2,772	291
Cowpea	16	894	85	999	434		535	947

Type	Zarqa market		Irbid market		Amman market		Central markets	
	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price
Cabbage	1,940	143	22,524	129	25,201	117	49,665	130
Red cabbage			139	276	1,030		1,169	276
Molokhia	1,161	280	8,911	219	10,547	176	20,619	225
Sagebrush	8	681			756		764	681
Minit	542	81	878	439	2,423		3,843	260
Grape leaves	157	2,437	173	2,161	1,818		2,148	2,299
butternut pumpkin	1	650	158	665	550		709	657

Table 14 Wholesale quantities and expected prices for imported fruits to the central markets

Type	Zarqa market		Irbid market		Amman market		Central markets	
	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price
Navel orange	805	555	5,763	624	19,674	694	26,242	624
French orange	96	413	273	349	1,294		1,663	381
grapefruit	245	281	203	335	2,316		2,764	308
Mendlina	121	446	1,071	367	4,239	96	5,431	303

Type	Zarqa market		Irbid market		Amman market		Central markets	
	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price
Shamouti orange	163	466	474	532	1,876		2,513	499
Calamondins	405	432	5,767	412	12,825	184	18,997	342
Lemon	1,195	766	8,279	728	11,496	1,040	20,970	845
Valencia orange	370	451	2,426	465	12,594	375	15,390	430
Blood orange	5	440	3,954	392	3,320	111	7,279	314
Pomelo	190	471	296	435	2,636	278	3,122	394
apricot	106	737	925	760	4,816	831	5,847	776
Loquats	58	1,216	164	1,126	861		1,083	1,171
Pear	287	1,005	1,148	1,004	5,800	1,580	7,235	1,197
cactus	31	461	50	342	1,402		1,483	402
strawberry	115	1,096	786	1,150	3,361	364	4,262	870
Grapes	5	1,560	92	1,266	115		212	1,413
Green almond	66	1,366	141	1,723	1,389		1,596	1,544
peach	144	675	1,858	566	3,880	57	5,882	433
persica	389	440	6,528	556	20,290	814	27,207	603
Nectarine	261	334	1,217	511	5,643		7,121	423
Apple	1,228	698	13,816	823	43,336	989	58,380	837
King orange	92	354	36	338	378		506	346

Type	Zarqa market		Irbid market		Amman market		Central markets	
	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price	Total quantities	Proposed price
pomegranate	295	732	2,001	718	7,228	262	9,524	571
Grapes	381	797	4,191	866	13,940	1,102	18,512	922
Melon	514	243	5,556	257	15,080	273	21,150	257
Watermelon	536	205	23,101	173	54,611	184	78,248	188
Figs	19	1,163	15	730	562		596	947
Dates	1	400	3	900	252		256	650
guava	129	964	899	1,355	4,443	216	5,471	845
banana	405	440	4,040	606	16,620	600	21,065	549
Green cherry	34	1,482	175	1,274	824	652	1,033	1,136

Table 15 Quantity and cost of raw materials as a production inputs

Required raw material	Quantity (Kg)	Cost per Kg (JD)	Total cost (JD)
Packaging Materials	2,340,000	0.05	117,000
Bottles	334,286	0.40	133,714
Cleaning materials	3,120	1.00	3,120
Preservatives and chemicals	3,120	2.00	6,240
Total			250,714

Required human resources

Grading and packing fruits and vegetables project requires 10 skilled workers in grading or 20 non-skilled workers working together in one shift, as well as a driver to supply the production requirements and to distribute the refrigerated products to the targeted markets, also the project needs an agricultural engineer specialized in post-harvest, or has good knowledge and experience in the preparation and grading and packing all kinds of vegetables, and has the ability to sell the product as well as bargaining on the purchase of production inputs of vegetables and fresh fruit from farmers or wholesale markets. There is a need to production supervisor for field supervision on the process ofproduction and packaging, and to a refrigerant technician for refrigerated stores ,and a skilled marketing agent in the distribution of production on the targeted markets and malls. The table below Shows the employment requirements

Table 16 required human resources

Job title	Number
Indirect staff	
Project Manager / Agricultural Engineer	1
Grading and packing supervisor	1
Refrigeration technician	1
Marketing Representative	1
Total	**4**
Direct staff	
Grading and packing workers	10
Loading and unloading workers	2
Driver	1
Total	**13**
Grand total	**17**

The following shows the job description for the required human resources:

- Agricultural Engineer/ Horticultural Production, Postharvest specialists: Full knowledge of how to process, pack and trade vegetables and fruits of all kinds, how to process them, sort and grading, and the ability to request and provide the necessary materials of production tools, supplies of vegetables, and supervise the manufacturing process during the preparation stage. Sorting, preparation, packaging, as well as adjusting the conditions of the production process according to instructions and scientific knowledge.

Sampling raw materials and conducting the laboratory tests. Control the production process and grading, packaging. Diagnosis of deviations from the standards, control the factors affecting product quality. Monitor the transportation, storage and cooling process, knowledge of the types of vegetables and fruits that are marketable. The products demanded and required in the markets, monitor the maturity and manufacturing process, control of physical, chemical and biochemical factors affecting manufacturing process, knowledge of the methods of preparation, packaging, packaging and types and forms, inspect the conformity of the product to required specifications, filling and work models, documenting expenses and proposing policies and action plans, Implementation, and evaluation of their results. The organization of work, distribution of workers among task and jobs, follow-up of their work in coordination with technical staff, suppliers and transport agencies to ensure the safety of the supply of raw materials and products.

- Production Supervisor: Receiving the different types of vegetables to be prepared, weight and inspection of quality. Sorting vegetables fruits and vegetables, storing vegetables by type, size or classes, isolating damaged items, peeling and cutting vegetables, removing limbs as required, control processing room staging and preparation, check machine readiness to work and clean it. Adjust the temperature, humidity, time and proper air rotation speed, run the sorting and grading machines, inspect the product's suitability to the required specifications, and weight and final product registration. Fill the product with appropriate packaging and weights, and save the final product in the appropriate storage rooms in terms of humidity, ventilation and temperature. Clean the vessels. Mobilization of business models. Applying occupational safety and health procedures and instructions.
- Sorting laborers Receiving the raw materials to be graded, weight the shipments. The development of raw materials within the system of sorting and grading, processing of tools and staging and packaging of packaging, scissors and knives. Receiving the vegetable crop, and unloading it in the designated place. Placing the packaging on the production line, sorting the deformed fruits, infected and small from healthy and mature fruits, and sorting the fruits according to the required specifications. Fill the fruit in the custom packaging and pack it according to instructions. Packing and packaging of containers, storage in warehouses or refrigerators until marketing them, and transporting containers to freight cars. Implementing occupational safety and health procedures and instructions.

- Marketing Representative: Following market developments, identifying and developing marketing objectives and sales methods, estimating expected sales volume, translating objectives into sales plans, and determining the expected sales volume for each product distribution market in different sales areas. Follow up implementation of policies and sales methods, and evaluation of promotional programs. Communicate with customers, make proposals to solve the problems they face, and follow up competitors and their sales programs to avoid future risks. Evaluate the results of sales, and assess the extent to which sales targets are achieved. Follow up the development of quality systems and standards. Preparing technical reports, providing occupational safety and health tools and means, providing appropriate health conditions at the workplace, Developing procedures and securing occupational safety and health requirements.

Location analysis

It is proposed that the site of the project in Jerash province, close to agricultural areas and can serve parts of Ajlun due to the close distance between them, where the project needs to collect fruits and vegetables from farmers or from the central markets when production is not available in nearby places. The project will succeed depending on the availability of raw materials (Vegetables and fruits) required for manufacturing processes wherever they could be found., such as Jarash and Ajloun area, due to the availability of the necessary infrastructure for the project, and easiness of marketing t, as well as the proximity of thr province to the high densely populated areas such as Amman, Irbid and Zarqa. Accordingly, it is proposed to establish the project in Jerash. Facility for processing, preparation, sorting and grading of vegetables and fruits so that the suitable location for production is chosen based on the availability of infrastructure services of water and electricity and the availability of raw materials,

Production and manufacturing processes description

Regardless of the size of the filling station, whether it is a small field filling unit, a small grading and filling central station , or a large plant, there are some criteria that must be taken into account when choosing the site, including design, construction and layout. Generally, the filling station operations include the following: product delivery; product grading; special procedures ; packing and packaging; and distribution.

The above processes requires support services and materials depending on the size of the process and the market. Certain specific procedures can be followed for some specific products that will be exported to known markets, but it is necessary to evaluate each case separately in many other cases. The period of grading and packaging works in stations may also be limited to the agricultural season. In this case, consideration should be taken to the possibility of using this center for other crops or findalternatives to cover operating costs.

Some products are harvested and traded in short or fast period, so that the quantities of crops can accumulate and can not be accomodatedby the center. In this case, there is an urgent need for refrigerated warehouses to prolong the circulation of such crops, and then the benefit is in operating the packing and grading station as well as prolonging the marketing period.

It is also important to keep in mind that the packing and grading centers are designed to operate continuously at the peak daily production during the season. The number and quality of machines and equipment are determined based on peak daily production during the season. Notice that that the determination of the capacity of grading machines and packaging has been increased compared to the actual production capacity in order to ensure efficient circulation of the production of crops, and producing high quality products without exhausting the machines and to save time for daily maintenance.

Freshly harvested crops have several processes to be prepared for marketing, shipping or storage, the most important are: preparation, basic sorting, cleaning, disease control, drying, industrial ripening, polishing, grading, packing, packaging, initial cooling. But it is not necessarily that all crops should pass all these processes, there are crops that do not need to be prepared and others do not need to ripen and so on. These operations are performed at the service center, where the products are received in the field packagings either by refrigerated or non-refrigerated vehicles. Then necessary processes for the crops are carried out from the initial washing, sorting, drying, sowing, scaling, gradation, filling and initial cooling according to the type and requirements of the crop. The products are then transported to markets or stored in refrigerated warehouses until they are transported to the market.

1. Preparation:

Some crops need to be placed in a shady place to lose some of their moisture or to protect them from sun such as dry onions that are harvested and placed in a shady place and covered with a vegetative part.

2. Initial sorting

The removal of infected fruits, diseases and insects that does not meet the standards of crop specifications such as color, shape, size and others. In this step using machinery may not be benefitical, but requires a well-trained workforce.

3. Cleaning :

The process of removing dirt, dust, soils and residues of spraying materials from the crop in a suitable manner for each crop, where the crop can be washed with water as in most vegetables crops or by wiping as in berries or using special brushes, as in peaches. It should be noted that the petals of flowers are affected if washed with water due to the existence of salts, so being careful is essential when washing the stems of flowers with water.

4. Disease control:

If the injury was in depth, there is no way to combat it except by using rays that cause damage to crops. If the injury is not in depth, gas evaporation such as sulfur dioxide, nitrogen oxides or decontamination can be used with chemical solutions such as sodium carbonate, borax, sodium hypochlorite and potassium permanganate, or by using heat, either by air or hot water.

Some hormones have been used to increase tissue immunity against diseases such as ethylene oxide D-2,4 , which has been used with lemons to prevent the infection of altenaria, while gerbilic acid with oranges and ethylene oxide on mango.

5. Drying

Drying is essential after washing or after disease controlling using solutions, to prevent the growth of pathogens due to excess moisture. Drying is usually carried out by passing hot air over the crop.

6. Industrial Maturation:

This process is used to deliver the crop good for consumption and homogenization maturity by using acetylene or ethylene gas in special rooms. Acetylene is less effective than ethylene about 100 times but easy to prepare by adding water to calcium carbide. Ethylene is the most common where banana, hazel, mango, pear and khaki can be cooked using 10 microliters / liter.

7. Polishing

This is done by placing a layer of wax on the fruits, either by brushing, dipping or painting it by brush as in apples, pears and cucumbers. The wax layer gives a gloss to the crop and also preserves the moisture of the fruit and isolates its surface from the microbes. Usually is the paraffin wax or karnoba are used in this process.

8. Grading

It is a process of sorting according to specific specifications and standardss such as color, size, weight, length, number of flowers in a bunch, and other specifications. In this process, fruits that were damaged during the other processes will be excluded.. These days grading is done by special machines for that.

9. Packing

Where a certain number of fruits of the crop are tied in packages by special thin threads or the elastic thread used for money , previously a strip of palm fronds was used. Examples of crops that could be tied in packs, carnations and most leafy vegetables such as spinach, parsley, parsley and some small fruit fruits that grow in the form of clusters such as rambutan.

10. Packaging

Good packaging ensures the protection of the crop and makes it attractive and economical. Packets are made of different materials. There are containers of wood, cardboard, plastic, flint or transparent polyethylene, and the packing may be in netting bags of canvas or plastic, as in Abu Farwa, onion, lemon or transparent plastic bags as as in the carrots. Transparent polyethylene gives extra attraction where the fruits can be seen from outside the packaging especially if they are properly packed.

Crop packing is important to make the product more attractive. For each crop, a certain method of packing. For example, apples are packed in corrugated cardboard in which mutual cells and pears are arranged either in a cross way or in rows , for large peaches deposited in plastic cells in a reciprocal manner. It may be placed pieces of corrugated or plain cardboard or scraps of paper in between fruits or layers of fruit to increase protection and moisture absorption.

11. Initial cooling:

This process, despite its importance is ignored by many farmers. Which is the rapid removal of crop heat before shipment or storage. This heat is known as the potential heat of the field. The internal temperature of some crops in high heat days is found to be higher than the air temperature. For example, it was found that Peach temperature on a hot day was higher than the temperature of the air by about 11 degrees Celsius. Cooling process slows maturation, reduces pathogenic activity, respiratory rate, water loss and facilitates final cooling. Initial cooling is important for many fruits and vegetables and is especially important for cutting flowers and fast-paced crops such as lettuce, grapes, strawberries, plums, apricots, tomatoes and others. Cooling is done by air, cold water, ice or discharge. Discharge, the air is drawn around the crop, reducing the pressure, and the boiling point becomes 0C ^{0} at pressure of 0,809psi(pound per square Inch), resulting in rapid evaporation that pulls the heat out of the crop.

12. Refrigerated transport:

Refrigerated cars are required to transport the product from the stores to the targeted markets so that the product is arranged in cartons or large containers to be placed on a wooden pallet or metal net to be loaded and unloaded using manual or mechanical cranes to ensure that the products are not damaged during these processes.

Technical requirements analysis

- Processing fresh vegetables by removing any leaves and parts that aren't required and washed well.
- Cut the vegetables according to the sizes required for each kind to become half cooked.
- Packaging in polyethylene bags, paper bags or paper or cork plates
- Packed containers are transported to the refrigeration room to be kept there until distribution.

Requirements include:

Main machinery and auxiliary equipment.

- Initial cooling units.
- Primary sorting and chopping units
- Grading units.
- Hygiene units.
- Packing units and it's accessories.
- Packaging units and it's accessories.
- Refrigeration units and it's accessories.

Land and construction works

The project requires 5,000 m2 for the the construction of freezing establishment and required 840m^2 for buildings and hangars to build the center..

Table17 required land

The land	Cost (JD/m2) deflation	Area (m2)
Required land	20	5,000

The following table shows the required construction works:

Table 18 required construction works

Construction works	Cost(JD/m2)	Area(m2)
Administrative offices	120	80
Raw material storage	100	100
Refrigerated warehouses	100	100
laboratory	30	30
Grading, packing workshop	120	500
Worker's Bathrooms and restroom		30
Total		**840**

Machinery and equipment

The needed machines consist mainly of cleaning and washing machine, convyers, tables for sorting, accessories of packing machines . In addition to the supplies of weighting, packing and filling machines, and refrigerated storges.

The following table shows required machinery and equipment:

Table 19 required machinery and equipment

Machinery and equipment	Qty	Unit cost(JD)
Vegetable and fruits processing and washing unit	1	4,000
Conveyor belts	3	2,500
Field Containers (Plastic)	500	1.5
Semi-automatic grading and packing lines	1	10,000

Manual grading and packing lines	3	8,000
Maturation and sterilization units	1	1,500
fresh vegetables packing machine	1	7,000
Belt conveyer	3	2,500
Primary cooling unit for products	1	10,000
Plain and wicker workbenches	5	250
Scales with laboratory equipments	1	1,300
Forklift and pallets	1	250
Mechanical and electrical tools with installation	1	2,000
Weighing machine (up to 100kg)	1	200
Sensitive Balance	2	250
Packaging and Printing Unit	1	500
Packing and deflation machine	1	1,000
Others	1	2,000

Furniture

The project needs the following furniture:

Table 20 required furniture for the project

Required furniture	Qty	Unit cost (JD)
Office furniture	2	2,000
Computer and other equipments	1	1,000
Kitchen equipments	1	1,500

Vehicles and transportaions

The following tables shows required vehicles and transportation means for the project:

Table 21 required vehicles

Required Vehicles	Qty	Unit cost(JD)
Refrigerated vehicle	1	25,000
Medium vehicle	1	15,000

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Icome tax on the net project income is 0 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 22 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 23 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 24 working time assumptions

Working time	
No. of weeks	52
No. of days per week	6
No. of days annually	312
No. of working hours	8

Table 25 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 26 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	0.00%
Utilities Expenses from Total Revenues	5.00%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	2.00%
Insurance Expenses from Total Assets Value	1.00%
Marketing Expenses from Total Revenues	3.00%

Table 27 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	100%
Income Tax after Deduction	0%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	**value**
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 28 Risk Premuim

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	7.50%
(3) Market Risk Premuim	9.93%
Income Tax Rate	0.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 29 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premuim	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	369,543
Loans Value	369,543
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50
Loans Value	369,543
Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	369,543
Equety Percentage	%50
Gross	
Project Value	739,085
Weighted Average Cost of Capital	**%11.97**

Capital Expenditures

The estimated cost of the project is 458,463 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 30 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost(JD)
Required Land	20	5,000	100,000
Total Cost			**100,000**

Table 31 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross / Cost
Administrative offices	120	80	9,600
Raw material storage	100	100	10,000
Refrigerated warehouses	100	100	10,000
laboratory	30	30	900
Grading, packing workshop	120	500	60,000
Worker's Bathrooms and restroom	100	30	3,000
Other	-		9,050
Total		**840**	**102,550**

Table 32 Machinery and equipment Capital Expenditure

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Vegetable and fruits processing and washing unit	6	4,000	4,000
Conveyor belts	3	2,500	7,500
Field Containers (Plastic)	500	1.5	750
Semi-automatic grading and packing lines	1	10,000	10,000
Manual grading and packing lines	3	8,000	24,000
Maturation and sterilization units	1	1,500	1,500
fresh vegetables packing machine	1	7,000	7,000
Belt conveyer	3	2,500	7,500
Primary cooling unit for products	1	10,000	10,000
Plain and wicker workbenches	5	250	1,250
Scales with laboratory equipments	1	1,300	1,300
Forklift and pallets	1	250	250
Mechanical and electrical tools with installation	1	2,000	2,000
Weighting machine (up to 100kg)	1	200	200
Sensitive Balance	2	250	500

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Packaging and Printing Unit	1	500	500
Packing and deflation machine	1	1,000	1,000
Others	1	2,000	2,000
Total			**81,250**

Table 33 Furniture Capital Expenditure

Furniture	Number	Unit cost (JD)	Total cost (JD)
Office furniture	2	2,000	4,000
Computer and its accessories	1	1,000	1,000
Kitchen and cafeteria equipment	1	1,500	1,500
Total			6,500

Table 34 Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Medium vehicle	1	15,000	15,000
Cooled vehicle	1	25,000	25,000
Total cost	**2**		**40,000**

Table 35 Pre-Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	2,000
Transportation	1,000
Marketing	2,000
Training	2,000
Total cost	**7,000**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 36 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	1	%8	108,216
General and administrative expenses	3	%25	900
Marketing expenses	3	%25	3,246
Indirect Salaries	3	%25	8,800
Total			**121,163**

Table 37 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	100,000	10%	110,000	%21.8
Construction works	102,550	10%	112,805	%22.4
Machinery &Equipment	81,250	10%	89,375	%17.7
Furniture	6,500	10%	7,150	%1.4
Vehicles	40,000	10%	44,000	%8.7
Pre-Operating Expenses	7,000	10%	7,700	%1.5
Working capital	121,163	10%	133,279	%26.4
Total (JD)	**458,463**		**504,309**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (504,309) JD, where 50% of the project will be financed through loans (252,155) JD, and the remaining (50%) through (Equity) with (252,155) JD. The following table summarizes the general structure of the required financing:

Table 38 Sources of Funding

Funding Sources	Amount	Percentage
Equity	252,155	%50
Loans	252,155	%50
Total	**504,309**	**%100**
Use of Fund	**Amount**	**Percentage**
Capital Expenditures	363,330	%72.05
Pre-Operating Expenses	7,700	%1.53
Working Capital	133,279	%26.43
Total	**504,309**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table 39 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	1,042,387	2,084,774	2,126,469	2,168,999	2,212,379	2,256,626	2,301,759	2,347,794	2,394,750	2,442,645
Utilities from Revenues	125,357	250,714	255,729	260,843	266,060	271,381	276,809	282,345	287,992	293,752
Service benefit expenses from revenues	7,605	15,514	15,824	16,141	16,464	16,793	17,129	17,472	17,821	18,177
Maintenance from Revenues	7,605	15,514	15,824	16,141	16,464	16,793	17,129	17,472	17,821	18,177
Disposables from Revenues	7,605	15,514	15,824	16,141	16,464	16,793	17,129	17,472	17,821	18,177
Direct Salaries	46,200	63,448	64,717	66,011	67,332	68,678	70,052	71,453	72,882	74,339
Others	61,838	122,274	124,719	127,214	129,758	132,353	135,000	137,700	140,454	143,263
Total	**1,298,597**	**2,567,753**	**2,619,108**	**2,671,490**	**2,724,920**	**2,779,418**	**2,835,007**	**2,891,707**	**2,949,541**	**3,008,532**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 40 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	500	510	520	531	541	552	563	574	586	598
Hospitality	500	510	520	531	541	552	563	574	586	598
Staionary and Printing	100	102	104	106	108	110	113	115	117	120
Consulting expenses	500	510	520	531	541	552	563	574	586	598
Training	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Insurance	500	510	520	531	541	552	563	574	586	598
Miscellaneous	500	316	323	329	336	342	349	356	363	370
Total	**3,600**	**3,478**	**3,548**	**3,619**	**3,691**	**3,765**	**3,840**	**3,917**	**3,995**	**4,075**

Marketing Expenses

Table 41 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	**12,986**	**25,678**	**26,191**	**26,715**	**27,249**	**27,794**	**28,350**	**28,917**	**29,495**	**30,085**

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 42 expected number of employees

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager / Agricultural Engineer	1	1	1	1	1	1	1	1	1	1
Grading and packing supervisor	1	1	1	1	1	1	1	1	1	1
Refrigeration technician	1	1	1	1	1	1	1	1	1	1
Marketing Representative	1	1	1	1	1	1	1	1	1	1
Total number of indirect employees	**4**	**4**	**4**	**4**	**4**	**4**	**4**	**4**	**4**	**4**
direct employees										
Grading and packing workers	10	10	10	10	10	10	10	10	10	10
Loading and unloading workers	2	2	2	2	2	2	2	2	2	2
Driver	1	1	1	1	1	1	1	1	1	1
Total number of direct employees	**13**	**13**	**13**	**13**	**13**	**13**	**13**	**13**	**13**	**13**
Grand total	**17**	**17**	**17**	**17**	**17**	**17**	**17**	**17**	**17**	**17**

Table 43 Expected monthly Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager / Agricultural Engineer	700	721	743	765	788	811	836	861	887	913
Grading and packing supervisor	500	515	530	546	563	580	597	615	633	652
Refrigeration technician	500	515	530	546	563	580	597	615	633	652
Marketing Representative	500	515	530	546	563	580	597	615	633	652
direct employees										
Grading and packing workers	300	309	318	328	338	348	358	369	380	391
Loading and unloading workers	250	258	265	273	281	290	299	307	317	326
Driver	350	361	371	382	394	406	418	430	443	457

Table 44 Expected Annual Salaries and Wages (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager / Agricultural Engineer	11,200	11,536	11,882	12,239	12,606	12,984	13,373	13,775	14,188	14,613
Grading and packing supervisor	8,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Refrigeration technician	8,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Marketing Representative	8,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total indirect salaries	**35,200**	**36,256**	**37,344**	**38,464**	**39,618**	**40,806**	**42,031**	**43,292**	**44,590**	**45,928**
direct employees										
Grading and packing workers	36,000	49,440	50,923	52,451	54,024	55,645	57,315	59,034	60,805	62,629
Loading and unloading workers	6,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Driver	4,200	5,768	5,941	6,119	6,303	6,492	6,687	6,887	7,094	7,307
Total direct salaries	**46,200**	**63,448**	**65,351**	**67,312**	**69,331**	**71,411**	**73,554**	**75,760**	**78,033**	**80,374**
Grand total	**81,400**	**99,704**	**102,695**	**105,776**	**108,949**	**112,218**	**115,584**	**119,052**	**122,623**	**126,302**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Table 45 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	110,000	%0.0	%0.0
Construction Works	112,805	%5.0	%0.0
Machinaries	89,375	%10.0	%2.0
Furniture	7,150	%10.0	%5.0
Vehicles	44,000	%10.0	%0.0
	363,330		

Table 46 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	363,330	365,475	367,674	369,927	372,238	374,608	377,037	379,530	382,086	384,708
Total Depreciation Expenses	19,693	19,907	20,127	20,352	20,584	20,821	21,063	21,313	21,568	21,831
Total Accumulated Depreciation	19,693	39,600	59,727	80,080	100,663	121,484	142,547	163,860	185,428	207,259
Total Additions	0	2,145	2,199	2,254	2,311	2,369	2,430	2,492	2,556	2,623
Total Net Book Values	343,637	325,875	307,947	289,848	271,575	253,124	234,490	215,670	196,658	177,450

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 47 Loan Details

Loan Value	193,362
Annual Interst Rate	9.50%
Loans Period	5
Grace Period	0
Loan Starts at year:	2018

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				252,155
2019	65,670	23,955	41,716	210,439
2020	65,670	19,992	45,679	164,761
2021	65,670	15,652	50,018	114,743
2022	65,670	10,901	54,770	59,973
2023	65,670	5,697	59,973	0

Income Statement

Table 48 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	1,521,000	3,102,840	3,164,897	3,228,195	3,292,759	3,358,614	3,425,786	3,494,302	3,564,188	3,635,472
Gross Operating Revenues	1,521,000	3,102,840	3,164,897	3,228,195	3,292,759	3,358,614	3,425,786	3,494,302	3,564,188	3,635,472
Operating Expenses	(1,298,597)	(2,567,753)	(2,619,108)	(2,671,490)	(2,724,920)	(2,779,418)	(2,835,007)	(2,891,707)	(2,949,541)	(3,008,532)
Gross Operating Profit	222,403	535,087	545,789	556,705	567,839	579,196	590,780	602,595	614,647	626,940
Gross Profit Percentage	*%15*	*%17*	*%17*	*%17*	*%17*	*%17*	*%17*	*%17*	*%17*	*%17*
Salaries and Benefits	(35,200)	(36,256)	(37,344)	(38,464)	(39,618)	(40,806)	(42,031)	(43,292)	(44,590)	(45,928)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
General and Administraive	(3,600)	(3,478)	(3,548)	(3,619)	(3,691)	(3,765)	(3,840)	(3,917)	(3,995)	(4,075)
Markeing Expenses	(12,986)	(25,678)	(26,191)	(26,715)	(27,249)	(27,794)	(28,350)	(28,917)	(29,495)	(30,085)
Pre Operating Expenses	(7,700)									
Gross Indirect Expenses	(59,486)	(65,412)	(67,083)	(68,798)	(70,558)	(72,366)	(74,221)	(76,126)	(78,081)	(80,089)
Income before Interest,	162,917	469,675	478,706	487,907	497,281	506,830	516,559	526,469	536,566	546,851
Fixed Assets Depreciations	(19,693)	(19,907)	(20,127)	(20,352)	(20,584)	(20,821)	(21,063)	(21,313)	(21,568)	(21,831)
Income before Tax and	143,224	449,768	458,579	467,555	476,697	486,010	495,495	505,157	514,998	525,021
Bank Interests	(23,955)	(19,992)	(15,652)	(10,901)	(5,697)	0	0	0	0	0
Income Before Tax	119,270	429,777	442,927	456,654	471,000	486,010	495,495	505,157	514,998	525,021
Income Tax	0	0	0	0	0	0	0	0	0	0
Net Profit	119,270	429,777	442,927	456,654	471,000	486,010	495,495	505,157	514,998	525,021
Net Profit Percentage	*%8*	*%14*	*%14*	*%14*	*%14*	*%14*	*%14*	*%14*	*%14*	*%14*
Compulsory Reserves	(11,927)	(42,978)	(44,293)	(45,665)	(47,100)	(48,601)	(49,550)	(50,516)	(51,500)	(52,502)
Retained Earnings	107,343	494,141	892,776	1,303,764	1,727,664	2,165,073	2,611,018	3,065,659	3,529,157	4,001,676

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 49 Expected Cashflows

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	119,270	429,777	442,927	456,654	471,000	486,010	495,495	505,157	514,998	525,021
Bank Interests	23,955	19,992	15,652	10,901	5,697	(0)	(0)	(0)	(0)	(0)
Depreciation	19,693	19,907	20,127	20,352	20,584	20,821	21,063	21,313	21,568	21,831
Total Operating Cashflows before Additions to Working Capital	162,917	469,675	478,706	487,907	497,281	506,830	516,559	526,469	536,566	546,851
Inventory (Increase/Decrease)	(86,866)	(86,866)	(3,475)	(3,544)	(3,615)	(3,687)	(3,761)	(3,836)	(3,913)	(3,991)
Accounts Receiviables (Increase/Decrease)	(126,750)	(131,820)	(5,171)	(5,275)	(5,380)	(5,488)	(5,598)	(5,710)	(5,824)	(5,940)
Accounts Paybales (Increase/Decrease)	21,351	18,897	805	821	837	854	871	889	907	925
Working Capital (Increase/Decrease)	(192,265)	(199,788)	(7,841)	(7,998)	(8,158)	(8,321)	(8,487)	(8,657)	(8,830)	(9,007)
Net Cashflows from Operating Activities	(29,348)	269,887	470,865	479,909	489,123	498,509	508,071	517,812	527,736	537,844
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(363,330)	(2,145)	(2,199)	(2,254)	(2,311)	(2,369)	(2,430)	(2,492)	(2,556)	(2,623)
Net Cashflows from Investments Activities	(363,330)	(2,145)	(2,199)	(2,254)	(2,311)	(2,369)	(2,430)	(2,492)	(2,556)	(2,623)
Cashflows from Financing Activities										
Capital	252,155									
Loan Amortization	(41,716)	(45,679)	(50,018)	(54,770)	(59,973)	(0)	(0)	(0)	(0)	(0)
Bank Interest Rate	(23,955)	(19,992)	(15,652)	(10,901)	(5,697)	0	0	0	0	0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Loans	252,155	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	438,639	(65,670)	(65,670)	(65,670)	(65,670)	0	0	0	0	0
Net (Increase/Decrease) in Cash	45,961	202,072	402,996	411,985	421,142	496,140	505,641	515,320	525,179	535,222
Cashflows at the Beginning of Period	0	45,961	248,033	651,030	1,063,015	1,484,157	1,980,296	2,485,938	3,001,258	3,526,437
Cashflows at the End of Period	45,961	248,033	651,030	1,063,015	1,484,157	1,980,296	2,485,938	3,001,258	3,526,437	4,061,659

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 50 Expected balance sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	45,961	248,033	651,030	1,063,015	1,484,157	1,980,296	2,485,938	3,001,258	3,526,437	4,061,659
Inventory	86,866	173,731	177,206	180,750	184,365	188,052	191,813	195,650	199,562	203,554
Accounts Receivable	126,750	258,570	263,741	269,016	274,397	279,884	285,482	291,192	297,016	302,956
Total current Assets	**259,577**	**680,334**	**1,091,977**	**1,512,781**	**1,942,918**	**2,448,233**	**2,963,233**	**3,488,099**	**4,023,015**	**4,568,169**

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non Current Assets										
Fixed Assets (net)	343,637	325,875	307,947	289,848	271,575	253,124	234,490	215,670	196,658	177,450
Total Non Current Assets	**343,637**	**325,875**	**307,947**	**289,848**	**271,575**	**253,124**	**234,490**	**215,670**	**196,658**	**177,450**
Total Assets	**603,214**	**1,006,209**	**1,399,923**	**1,802,629**	**2,214,493**	**2,701,357**	**3,197,723**	**3,703,769**	**4,219,673**	**4,745,618**
Liabilities	-	-	-	-	-					
Current Liabilities										
Paybles	21,351	40,248	41,053	41,874	42,712	43,566	44,437	45,326	46,233	47,157
Remaining amount of Loan	45,679	50,018	54,770	59,973	0	0	0	0	0	0
Total current Liabilities	**67,029**	**90,266**	**95,823**	**101,847**	**42,712**	**43,566**	**44,437**	**45,326**	**46,233**	**47,157**
Non Current Liabilities										
Long Terms Loans	164,761	114,743	59,973	(0)	(0)	(0)	(0)	(0)	(0)	(0)
Total Long Term Liabilities	**164,761**	**114,743**	**59,973**	**(0)**	**(0)**	**(0)**	**(0)**	**(0)**	**(0)**	**(0)**
Total Liabilities	**231,790**	**205,009**	**155,796**	**101,847**	**42,712**	**43,566**	**44,437**	**45,326**	**46,233**	**47,157**
Owners Equity										
Shareholders Contributions	252,155	252,155	252,155	252,155	252,155	252,155	252,155	252,155	252,155	252,155
Statutory Reseve	11,927	54,905	99,197	144,863	191,963	240,564	290,113	340,629	392,129	444,631
Retained Profits	107,343	494,141	892,776	1,303,764	1,727,664	2,165,073	2,611,018	3,065,659	3,529,157	4,001,676
Total Equity	371,424	801,201	1,244,128	1,700,782	2,171,781	2,657,791	3,153,286	3,658,443	4,173,440	4,698,461
Total Liabilities and Equity	**603,214**	**1,006,209**	**1,399,923**	**1,802,629**	**2,214,493**	**2,701,357**	**3,197,723**	**3,703,769**	**4,219,673**	**4,745,618**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 51 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(504,309)	(21,648)	267,742	468,667	477,655	486,812	496,140	505,641	515,320	525,179	535,222	6,988,459
Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash Flows	**(504,309)**	**(19,334)**	**213,576**	**333,901**	**303,938**	**276,662**	**251,832**	**229,227**	**208,650**	**189,918**	**172,867**	**2,257,140**

The table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 52 Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(504,309)	(21,648)	267,742	468,667	477,655	486,812	496,140	505,641	515,320	525,179	535,222
Project Value Returned	504,309	525,957	258,215	(210,452)	(688,107)	(1,174,919)	(1,671,059)	(2,176,700)	(2,692,021)	(3,217,200)	(3,752,422)
Payback Period (Year)	2	1	1	0	0	0	0	0	0	0	0

Table 53 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.97
Net Present Value for Cashflows	3,914,068
Payback Period	2
Internal Rate of Return	%55.26

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 54 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC	Sensitivity Analysis
Original Scenario	55.26%	2	3,914	11.97%
Revenues declined by 10%	22.27%	6	744	11.97%
Operating Expenses Increased by 10%	27.15%	5	1,171	11.97%
Total cost Increased by 10%	25.75%	5	1,120	11.97%
Total cost Increased and revenues decreased by 10%	< 0	10	-2,045	11.97%

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 3,914,068 Jordan Dinars considering that the project provides 17 Job Opportunitites for the governorate residents.